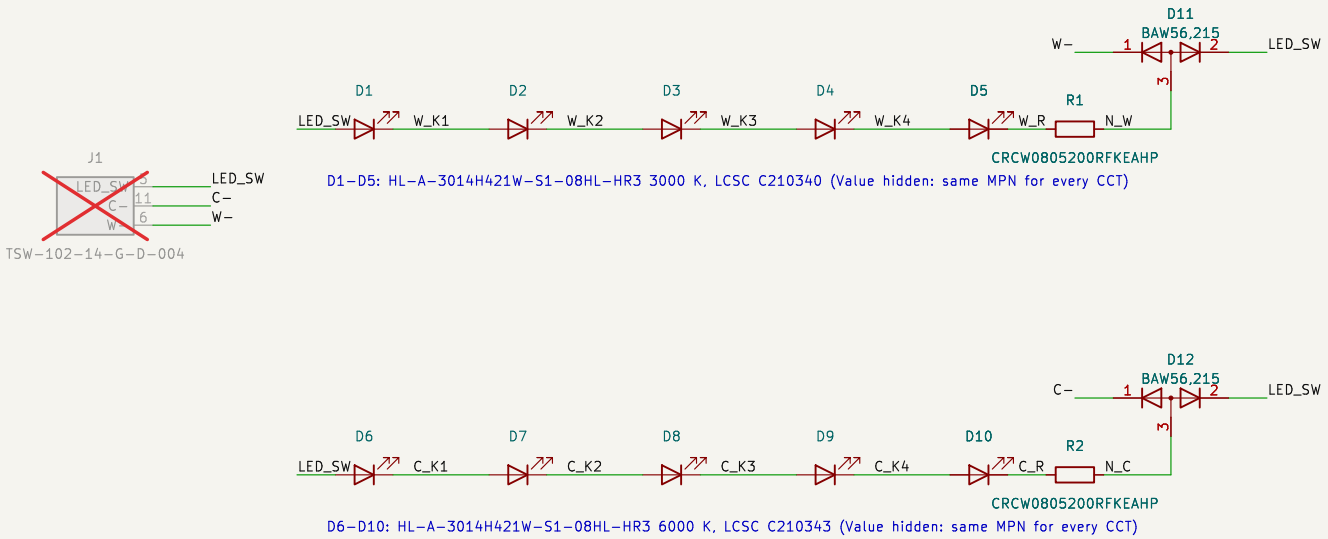



D1–D5: HL–A–3014H421W–S1–08HL–HR3 3000 K, LCSC C210340 (Value hidden: same MPN for every CCT)

D6–D10: HL–A–3014H421W–S1–08HL–HR3 6000 K, LCSC C210343 (Value hidden: same MPN for every CCT)

HOW IT CONNECTS
J1 (Samtec TSW–102–14–G–D–004) plugs into the main board's J6 through the case's top–wall opening.
J1 is DNP at JLC (hand–soldered by the builder): TSW–102–14–G–D, DigiKey 612–TSW–102–14–G–D–ND, position 4 pushed out.
Pins are numbered by the J6 pin they mate: 5 = LED_SW (anode bus), 6 = W– (warm return), 11 = C– (cool return).
J6–12 (P+_FUSE, battery) has NO pin and no hole; no GND/3V3 or other J6 net exists on this board.
Samtec pos 1 = J6–5, pos 2 = J6–11, pos 3 = J6–6, pos 4 (omitted) = J6–12.

EACH STRING: LED_SW –> 5 LEDs –> Rs 200 R (CRCW0805200RFKEAHP) –> node N –> BAW56 pin 3 (CA); pin 1 (K1) –> W–/C– (series block);
pin 2 (K2) –> LED_SW (clamp: holds the string's reverse voltage to one diode drop if the plug is misaligned).
Driver: constant current 13.3 mA (14.0 max), one string at a time; OVP 24.25 V min; design ceiling 22.0 V.
String at 13.3 mA: 5 x Vf 13.0..15.8 V + Rs 2.66 V + BAW56 0.60..0.95 V = LED_SW 16.6..20.4 V (typ 18.0 V) < 22.0 V.
Hot–plug peak through Rs 200 R: TPS923610 44..49 mA, TPS923611 68..74 mA < IFP 80 mA.

LEDs: HONGLITRONIC HL–A–3014H421W–S1–08HL–HR3; the MPN does not encode CCT: warm D1–D5 = LCSC C210340 (3000 K),
cool D6–D10 = LCSC C210343 (6000 K). Pad 1 (large) = cathode. Diode–test one loose LED before assembly.

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Clip–on warm/cool reading light for the Silkscreen reader; plugs into main–board J6
**idc LLC**

Sheet: /
File: silkscreen–light–attachment.kicad_sch

***Title: Silkscreen light attachment***

| Size: A4 | Date: 2026–09–25 | **Rev: A** |
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| KiCad E.D.A. 9.0.6 | | Id: 1/1 |